DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

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Lecture 6: Log vs. Power Utility

CRRA · CARA · Which Utility Should You Use?

August 8, 2026

Today's Agenda

1. Power (CRRA) utility – and log utility as its limit
2. Exponential (CARA) utility
3. Two exact scaling results (not just approximations)
4. Which one should you use? – a preview of Week 3

Learning Objectives

- Define CRRA utility; show it gives constant $R(W)$, decreasing $A(W)$.
- Show log utility is the $\gamma \to 1$ limit of CRRA.
- Define CARA utility; show it gives constant $A(W)$, increasing $R(W)$.
- State the exact wealth-scaling properties of each family's certainty equivalent.

Power (CRRA) Utility

Definition

$$u(W) = \frac{W^{1-\gamma} - 1}{1-\gamma}, \qquad \gamma > 0,\ \gamma \neq 1$$

Remark: why the "-1"?

Adding/subtracting a constant never changes $A(W)$ or $R(W)$. It's there only so $u(W)$ has a well-defined limit as $\gamma \to 1$.

Risk Aversion Under Power Utility

$$u'(W) = W^{-\gamma}, \qquad u''(W) = -\gamma W^{-\gamma-1}$$

Risk Aversion Under Power Utility

$$A(W) = \frac{\gamma}{W} \qquad R(W) = \gamma$$

$R(W) = \gamma$ constant $\Rightarrow$ **CRRA**. $A(W) = \gamma / W$ decreasing $\Rightarrow$ DARA, automatically!

A Photocopier for Decisions

Scale Invariance

A CRRA investor's decision about a gamble that's a fixed *percentage* of wealth is identical at any wealth level. Feed the decision through a photocopier scaling wealth *and* gamble by the same factor – it comes out unchanged. That's exactly why γ is called a *relative* risk-aversion parameter.

Log Utility as the Limit $\gamma \to 1$

At $\gamma = 1$: $\dfrac{W^{1-\gamma} - 1}{1-\gamma} \to \dfrac{0}{0}$. L'Hôpital:

$$\lim_{\gamma \to 1} \frac{W^{1-\gamma} - 1}{1-\gamma} = \lim_{\gamma \to 1} \frac{-W^{1-\gamma} \ln W}{-1} = \ln W$$

Log Utility

$u(W) = \ln(W)$ is exactly the $\gamma = 1$ member of CRRA: $A(W) = 1/W$, $R(W) = 1$.

Bernoulli's 1738 fix for the St. Petersburg Paradox – now revealed as $\gamma = 1$ CRRA.

Exact Result: CRRA Scales Linearly with Wealth

Proportional gamble $W = \overline{W}(1+\tilde{\varepsilon})$, $\tilde{\varepsilon}$ independent of $\overline{W}$:

$$\mathbb{E}[u(W)] = \ln \overline{W} + \mathbb{E}[\ln(1+\tilde{\varepsilon})] \;\Rightarrow\; CE = \overline{W} \cdot e^{\mathbb{E}[\ln(1+\tilde{\varepsilon})]}$$

$\pi / \overline{W}$ is **constant** – exact, not approximate!

Verifying Exact Proportional Scaling

$u(W) = \ln(W)$, $\pm 10\%$ gamble on wealth

$\overline{W}$	CE	π	$\pi / \overline{W}$
Rs. 10,00,000	Rs. 9,94,987	Rs. 5,013	0.5013%
Rs. 1,00,00,000	Rs. 99,49,874	Rs. 50,126	0.5013%

Wealth $\times 10 \Rightarrow \pi \times 10$ exactly, while $\pi / \overline{W}$ stays frozen.

Exponential (CARA) Utility

Definition

$$u(W) = -\frac{1}{a}e^{-aW}, \qquad a > 0$$

$$u'(W) = e^{-aW}, \qquad u''(W) = -ae^{-aW}$$

Risk Aversion Under Exponential Utility

$$A(W) = a \text{ (constant!)} \qquad R(W) = aW$$

A Fixed-Size Cushion, Whatever Your Height

CARA's Signature Property

A CARA investor's willingness to risk a *fixed number of rupees* never changes – Rs. 10,000 or Rs. 10 crore, same cushion size. Convenient mathematically. But less realistic than CRRA: we expect richer people to risk *more* rupees, which is exactly the DARA property CARA lacks.

Exact Result: CARA Is Wealth-Independent

Additive gamble $W = \overline{W} + \varepsilon$, ε independent of $\overline{W}$:

$$\mathbb{E}[u(W)] = -\frac{1}{a} e^{-a\overline{W}} \, \mathbb{E}[e^{-a\varepsilon}] \;\Rightarrow\; CE = \overline{W} - \frac{1}{a} \ln \mathbb{E}[e^{-a\varepsilon}]$$

$$\pi = \frac{1}{a} \ln \mathbb{E}[e^{-a\varepsilon}] \quad \text{does } \textbf{not} \text{ depend on } \overline{W} \text{ at all.}$$

Verifying Exact Wealth-Independence

$a = 5 \times 10^{-7}$, gamble $\pm$Rs. 1,00,000

W	π
Rs. 10,00,000	Rs. 2,498.96
Rs. 1,00,00,000	Rs. 2,498.96
Rs. 10,00,00,000	Rs. 2,498.96

Wealth $\times 100$ – π doesn't move at all.

CRRA vs. CARA: Side by Side

	Power / Log (CRRA)	**Exponential (CARA)**
$A(W)$	γ / W (decreasing)	a (constant)
$R(W)$	γ (constant)	aW (increasing)
Exact scaling	π / W const. (proportional risk)	π const. (additive risk)
Realism	Generally good	Poor individually, but tractable
Typical use	Long-run asset pricing	Gaussian / tractable settings

Preview: This Choice Drives Next Week's Math

Remark

Week 3: how much wealth to put in a risky asset? Under **CRRA**: optimal *fraction* in the risky asset is wealth-independent. Under **CARA** (with normal returns): optimal *rupee amount* is wealth-independent.

Not a new idea – today's exact scaling results, applied to an optimization problem.

Key Takeaways

Key Takeaway

- **Power (CRRA)**: $R(W) = \gamma$ constant, $A(W) = \gamma / W$ decreasing (DARA free of charge).
- **Log utility**: exactly $\gamma \to 1$ of CRRA, $R(W) = 1$.
- **Exponential (CARA)**: $A(W) = a$ constant, $R(W) = aW$ increasing.
- Exact (not approximate): CRRA $\Rightarrow \pi$ scales linearly with wealth for proportional risk; CARA $\Rightarrow \pi$ is wealth-independent for additive risk.

Check Your Understanding

Before Week 3

1. Verify directly (no limit) that $u(W) = \ln W$ gives $A(W) = 1/W$, $R(W) = 1$.
2. CRRA investor, $\gamma = 3$: $\pi/\overline{W} = 1.2\%$ at $\overline{W} =$ Rs. 5,00,000. Find π at $\overline{W} =$ Rs. 50,00,000.
3. Why might a modeler pick CARA despite it being empirically less realistic?

Looking Ahead

Module 1 complete: TVM & yield curves (Week 1) + risk aversion (Week 2).

Portfolio Theory – Week 3

Mean-Variance Analysis · The Efficient Frontier · Two-Fund Separation